Necessary and Sufficient Conditions for State Feedback Equivalence to Negative Imaginary Systems

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***Abstract*—In this article, we present necessary and sufficient conditions under which a linear time-invariant (LTI) system is state feedback equivalent to a negative imaginary (NI) system. More precisely, we show that a minimal LTI strictly proper system can be rendered NI using full state feedback, if and only if it can be output transformed into a system, which has relative degree less than or equal to two and is weakly minimum phase. We also considered the problems of state feedback equivalence to output strictly negative imaginary systems and strongly strictly negative imaginary systems. Then, we apply the NI state feedback equivalence results to robustly stabilize an uncertain system with strictly negative imaginary uncertainty. A real-world example is provided to illustrate the proposed results, for the purpose of stabilizing an uncertain system.**

***Index Terms*—Controller synthesis, feedback equivalence, negative imaginary (NI) system, robust control, stabilization.**

I. Introduction

NEGATIVE imaginary (NI) systems theory was introduced in [1] and [2] and has attracted attention since then [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13]. Motivated by the control of flexible structures [14], [15], [16], NI systems theory has been applied in many fields including nanopositioning control [17], [18], [19], [20] and the control of lightly damped structures [7], [21], [22], etc. Typical mechanical NI systems are systems with colocated force actuators and position sensors. In this sense, NI systems theory provides an alternative to positive real (PR) systems theory [23], as PR systems theory uses negative velocity feedback control while NI systems theory uses positive position feedback control. In comparison with PR systems theory, one advantage of NI systems theory is that it allows systems to have relative degrees of zero, one, and two, while PR systems can only have relative degrees of zero and one.

Roughly speaking, a square transfer matrix is NI if it is stable and its Hermitian imaginary part is negative semidefinite for all frequencies $\omega \geq 0$. For a single-input single-output NI system, its frequency response has a phase lag between 0 to $-\pi$ radians for all frequencies $\omega > 0$. It is shown using a set of linear matrix inequalities (LMIs) in the NI lemma that a system is NI if it is dissipative, with the supply rate being the inner product of its input and the time derivative of its output [3], [4], [24]. If certain conditions are satisfied, the interconnection of an NI system $R(s)$ and a strictly negative imaginary (SNI) controller $G(s)$ is proved to be asymptotically stable in various versions (see [1], [5], and [25]). For example, If $R(s)$ is an NI system without poles at the origin satisfying $R(\infty) = 0$ and $G(s)$ is an SNI system, then the positive feedback interconnection of $R(s)$ and $G(s)$ is internally stable, if and only if $\lambda_{\max}(R(0)G(0)) < 1$; see [25]. A similar result also holds when $G(s)$ is an output strictly negative imaginary (OSNI) system (see [8] and [26] for OSNI systems).

The problem of rendering a system PR using state feedback control in order to achieve stabilization has been investigated in many papers (see [27], [28], etc). For example, [28] renders a linear system PR and this result is then generalized to nonlinear systems in [29] using passivity theory. Further nonlinear generalizations of these ideas are presented in the papers [30], [31], [32], [33]. In these papers, such PR or passivity state feedback equivalence results are then applied to stabilize systems with specific nonlinearities. One of the necessary and sufficient conditions for state feedback equivalence to a passive or PR system is that the original system must have relative degree one. This restriction stems from the nature of passivity and PR systems and, as a result, rules out a wide variety of control systems with relative degree two, such as mechanical systems with force actuators and position sensors. To overcome this limitation and to complement the existing results that are based on passivity and PR systems theory, we consider the problem of state feedback equivalence to NI systems.

In this article, we investigate necessary and sufficient conditions under which a linear system with the minimal realization $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ is state feedback equivalent to an NI system. Suppose the system has no zeros at the origin. We show that such a system can be made NI via the use of state feedback control, if and only if

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1) it can be output transformed into a system with relative degree less than or equal to two; and 2) the transformed system is weakly minimum phase (see e.g., [34] for details of the terminology in feedback stabilization). The idea of applying an output transformation comes from the fact that a system in general does not always have a relative degree vector and hence, does not always have a normal form. However, we show that the property of NI state feedback equivalence is invariant under a nonsingular output transformation because its effect can be compensated for by an additional input transformation. Moreover, we show that a system can be made OSNI, if and only if it can be made NI. In particular, we show that a system is state feedback equivalent to a strongly strictly negative imaginary (SSNI) system, if and only if it has a relative degree vector $\{1, \ldots, 1\}$ and is minimum phase. The proposed NI state feedback equivalence results are then applied to robustly stabilize an uncertain system with SNI uncertainty.

The contribution of this article is to provide conditions under which a system is state feedback equivalent to an NI system, an OSNI system, or an SSNI system. This work is the first in the literature where necessary and sufficient conditions for NI state feedback equivalence is provided for a general linear system. In the preliminary conference paper [35], we consider cases where a system has relative degree of either one or two, which rules out the case that a system has mixed relative degrees one and two. In this article, we consider the general case which allows the system to have mixed relative degrees. Also, the relative degree condition was an assumption in [35], while it is a part of the necessary and sufficient conditions in this article. This makes the present article a complete result for the NI state feedback equivalence problem.

This work is important because it provides a simple necessary and sufficient condition to determine if a given system can be made NI via the use of state feedback. This condition is stated as a relative degree and minimum phase condition. This article also contributes in providing a method to stabilize systems with SNI uncertainty by making the plant NI. The problem of making a system NI using state feedback was also considered in [2] and [36]. However, the NI state feedback control results in [2] and [36] were given in terms of the existence of solutions to LMIs or Riccati equations, which may be difficult to solve. Also, the present article provides necessary and sufficient conditions while [2] and [36] only provide sufficient conditions. In addition, this article provides explicit formulas and a procedure for the construction of the required state feedback control input. The NI state feedback equivalence problem was recently investigated for nonlinear input-affine systems in [37] and systems in a normal form with constrained nonlinearity in [38]. However, [37] only provided sufficient conditions for the equivalence, and the conditions in [38] were specific to a constrained normal form rather than a general nonlinear system model. Also, certain assumptions were required explicitly in [37] and implicitly in [38] to guarantee the existence of a specific normal form. These assumptions are more complex than the relative degree and minimum phase conditions provided in this work. Therefore, the nonlinear results in [37] and [38] did not specialize to the results in the present article in the case of linear systems.

The rest of this article is organized as follows. Section II provides the essential background on NI systems theory. Section III contains the main results of this article, where we derive necessary and sufficient conditions under which a system can be made NI or OSNI using state feedback control. Formulas for the required state feedback matrices are provided in the proofs. In Section IV, an SSNI state feedback equivalence result is also provided. Section V applies the NI state feedback equivalence results presented in Section III in stabilizing an uncertain system with SNI uncertainty. Section VI illustrates the presented results with a numerical example. Finally, Section VII concludes this article.

Notation: The notation in this article is standard. $\mathbb{R}$ and $\mathbb{C}$ denote the sets of real and complex numbers, respectively. $\mathbb{N}$ denotes the set of nonnegative integers. $j\mathbb{R}$ denotes the set of imaginary numbers. $\mathbb{R}^{m\times n}$ and $\mathbb{C}^{m\times n}$ denote the spaces of real and complex matrices of dimension $m \times n$, respectively. $\Re[\cdot]$ is the real part of a complex number. A^T and A^* denote the transpose and complex conjugate transpose of a matrix A, respectively. A^{-T} denotes the transpose of the inverse of A; i.e., $A^{-T} = (A^{-1})^T = (A^T)^{-1}$. $ker(A)$ denotes the kernel of A. $spec(A)$ denotes the spectrum of A. $\lambda_{\max}(A)$ denotes the largest eigenvalue of a matrix A with real spectrum. For a symmetric or Hermitian matrix P, $P > 0$ $(P \geq 0)$ denotes the property that the matrix P is positive definite (positive semidefinite) and $P < 0\,(P \leq 0)$ denotes the property that the matrix P is negative definite (negative semidefinite). For a positive definite matrix P, we denote by $P^{\frac{1}{2}}$, the unique positive definite square root of P. We denote the open left half plane of the complex plane by $\mathbb{C}^- = \{s | \Re[s] < 0\}$.

II. Preliminaries

The following definitions and lemmas are needed.

Definition 1: (Negative imaginary systems) [3]. A square real-rational proper transfer function matrix $R(s)$ is said to be negative imaginary if the following conditions hold.

1) $R(s)$ has no poles at the origin and in $\Re[s] > 0$.
2) $j[R(j\omega) - R^*(j\omega)] \geq 0$ for all $\omega \in (0, \infty)$ except for values of ω where $j\omega$ is a pole of $R(s)$.
3) If $j\omega_0$ with $\omega_0 \in (0, \infty)$ is a pole of $R(s)$, then it is a simple pole and the residue matrix $K_0 = \lim_{s\to j\omega_0}(s - j\omega_0)jR(s)$ is Hermitian and positive semidefinite.

Note that the NI systems definition has been generalized beyond Definition 1 to allow for systems with poles at the origin (see [5]). However, we only investigate conditions for state feedback equivalence to NI systems as per Definition 1 because: 1) NI systems satisfying Definition 1 is guaranteed to be Lyapunov stable under zero input while this is not true for NI systems with poles at the origin; and 2) many existing NI system results are only applicable to NI systems without poles at the origin. Therefore, it is of interest to make a system NI without poles at the origin.

Definition 2: (Strictly negative imaginary systems) [3]. A square real-rational proper transfer function matrix $R(s)$ is said to be strictly negative imaginary if the following conditions are satisfied.

1) $R(s)$ has no poles in $\Re[s] \geq 0$.
2) $j[R(j\omega) - R^*(j\omega)] > 0$ for all $\omega \in (0, \infty)$.

Definition 3: (Output strictly negative imaginary systems) [26]. A square real-rational proper transfer function matrix $R(s)$ is said to be output strictly negative imaginary if there exists a scalar $\epsilon > 0$ such that

$$j\omega[R(j\omega) - R(j\omega)^*] - \epsilon\omega^2 \bar{R}(j\omega)^* \bar{R}(j\omega) \geq 0$$

for all $\omega \in \mathbb{R} \cup \infty$ where $\bar{R}(j\omega) = R(j\omega) - R(\infty)$. In this case, we say $R(s)$ is OSNI with a level of output strictness ϵ.

Definition 4: (Strongly strictly negative imaginary systems) [39]. A square real-rational proper transfer function matrix $R(s)$ is said to be strongly strictly negative imaginary if the following conditions are satisfied.

1) $R(s)$ is SNI.
2) $\lim_{\omega\to\infty} j\omega[R(j\omega) - R^*(j\omega)] > 0$ and $\lim_{\omega\to 0} j\frac{1}{\omega}[R(j\omega) - R^*(j\omega)] > 0$.

Lemma 1: (NI Lemma) [1], [3] Let (A, B, C, D) be a minimal state-space realization of a $p \times p$ real-rational proper transfer function matrix $R(s)$ where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times p}$. Then $R(s)$ is NI if and only if the following conditions are satisfied.

1) $\det(A) \neq 0$, $D = D^T$.
2) There exists a matrix $Y = Y^T > 0$, $Y \in \mathbb{R}^{n\times n}$ such that

$$AY + YA^T \leq 0, \qquad \text{and} \qquad B + AYC^T = 0.$$

Lemma 2: (SSNI Lemma) [39]. Given a square transfer function matrix $R(s) \in \mathbb{R}^{p\times p}$ with a state-space realization (A, B, C, D), where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times p}$. Suppose $R(s) + R(-s)^T$ has normal rank p and (A, B, C, D) has no observable uncontrollable modes. Then, A is Hurwitz and $R(s)$ is SSNI, if and only if $D = D^T$ and there exists a matrix $Y = Y^T > 0$ that satisfies the following conditions:

$$AY + YA^T < 0, \quad \text{and} \quad B + AYC^T = 0.$$

Lemma 3: (OSNI Lemma) [26]. Let (A, B, C, D) be a minimal state-space realization of a $p \times p$ real-rational proper transfer function matrix $R(s)$, where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times p}$, $C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times p}$. Let $\epsilon > 0$ be a scalar. Then, $R(s)$ is OSNI with a level of output strictness ϵ, if and only if $D = D^T$ and there exists a matrix $Y = Y^T > 0$, $Y \in \mathbb{R}^{n\times n}$ such that

$$AY + YA^T + \epsilon(CAY)^T CAY \leq 0, \quad \text{and} \quad B + AYC^T = 0.$$

The following version of the NI stability theorem was published in [25] and is needed in our main robust stability result.

Lemma 4: (Internal stability of interconnected NI systems) [25]. Consider an NI transfer function matrix $R(s)$ satisfying $R(\infty) = 0$ and an SNI system $G(s)$. Then, the positive feedback interconnection $[R(s), G(s)]$ is internally stable, if and only if $\lambda_{\max}(R(0)G(0)) < 1$. (see [1] for the definition of internal stability and positive feedback interconnection.)

III. NI State Feedback Equivalence

In this section, we derive necessary and sufficient conditions under which a system is state feedback equivalent to an NI system and an OSNI system, respectively. Consider a system with the state-space model

$$\dot{x} = \mathcal{A}x + \mathcal{B}u, \tag{1a}$$

$$y = \mathcal{C}x \tag{1b}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^p$ is the input, and $y \in \mathbb{R}^p$ is the output. Here, $\mathcal{A} \in \mathbb{R}^{n\times n}$, $\mathcal{B} \in \mathbb{R}^{n\times p}$, and $\mathcal{C} \in \mathbb{R}^{p\times n}$. We assume that $rank(\mathcal{B}) = rank(\mathcal{C}) = p$.

Remark 1: In the system model (1) we investigate in this article, there is no input feedthrough term in the output equation; i.e., $\mathcal{D} = 0$. The addition of such a feedthrough term would considerably complicate the problem and is left for future research.

It will be shown later in the article that one essential condition for NI state feedback equivalence is on a system's relative degree. Since there are various versions of definitions of relative degree in the literature, we provide the following definitions for the system (1).

Definition 5: (see also [34] and [40]). A vector $r = \{r_1, \ldots, r_p\} \in \mathbb{N}^p$ is called the relative degree vector of system (1) if the following conditions are satisfied.

1) For all $i = 1, \ldots, p$

$$\begin{aligned} &\mathcal{C}_i\mathcal{A}^j\mathcal{B} = 0 \quad \text{for} \quad j = 0, \ldots, r_i - 2; \\ &\text{and} \quad H(r)_i := \mathcal{C}_i\mathcal{A}^{r_i - 1}\mathcal{B} \neq 0. \end{aligned} \tag{2}$$

2) $\det(H(r)) \neq 0$.

Here, $\mathcal{C}_i$ denotes the ith row of the matrix $\mathcal{C} \in \mathbb{R}^{p\times n}$ and

$$H(r) = \begin{bmatrix} \mathcal{C}_1\mathcal{A}^{r_1-1}\mathcal{B} \\ \vdots \\ \mathcal{C}_p\mathcal{A}^{r_p-1}\mathcal{B} \end{bmatrix}. \tag{3}$$

Condition 1 in this definition indicates that the ith output has its r_ith time derivative depending explicitly on the inputs.

As was explained in [40], in the case that (1) is a MIMO system; i.e., $p \geq 2$, Condition 2 in Definition 5 is not always satisfied. The components in the relative degree vector r are invariant under a nonsingular state transformation. However, a nonsingular output transformation can change the components in the vector r and in some cases transform the realization $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ to $(\mathcal{A}, \mathcal{B}, \tilde{\mathcal{C}})$, where $\tilde{\mathcal{C}} = T_y\mathcal{C}$, $T_y \in \mathbb{R}^{p\times p}$, and $\det(T_y) \neq 0$, which satisfies Condition 2 in Definition 5.

Note that in general there does not always exist an output transformation for a system of the form (1) such that the resulting system has a relative degree vector. In what follows, we need the notion of a leading incomplete relative degree vector [40].

Definition 6 ([40]): A vector $r = \{r_1, \ldots, r_p\} \in \mathbb{N}^p$ is called a leading incomplete relative degree (LIRD) vector of the system (1) if the following conditions are satisfied.

1) For all $i = 1, \ldots, p$

$$\begin{aligned} &\mathcal{C}_i\mathcal{A}^j\mathcal{B} = 0 \quad \text{for} \quad j = 0, \ldots, r_i - 2; \\ &\text{and} \quad \mathcal{C}_i\mathcal{A}^{r_i-1}\mathcal{B} \neq 0. \end{aligned}$$

2) $r_i \leq r_{i+1}$ for $i = 1, \ldots, p - 1$.
3) For any set of pairwise distinct indices $i_1, \ldots, i_q \in \{1, 2, \ldots, p\}$ such that $r_{i_1} = r_{i_2} = \cdots = r_{i_q}$, the rows

$H(r)_{i_1}, \ldots, H(r)_{i_q}$ are linearly independent, where $H(r)$ is defined in (3) and $H(r)_i$ is defined in (2).

As was explained in [40] and [41], if a LIRD vector is such that all rows in $H(r)$ are linearly independent, then this LIRD vector is a relative degree vector as defined in Definition 5. This relationship can also be observed by comparing Definitions 5 and 6.

Lemma 5 ([41]): For any controllable system with the realization $(\mathcal{A}, \mathcal{B}, \mathcal{C})$, there exists a nonsingular output transformation such that the resulting system has an LIRD vector.

Proof: This follows directly from Remark 4 and Lemma 4 in [41]. ∎

In this article, we consider systems whose relative degree vector only consists of numbers less than or equal to two, as we show later that this is one of the necessary conditions for state feedback equivalence to NI systems.

Definition 7: The system (1) is said to have relative degree less than or equal to two if it has a relative degree vector $r = \{r_1, \ldots, r_p\}$, where $1 \le r_i \le 2$ for all $i = 1, \ldots, p$.

A. State Feedback Equivalence to NI Systems

In this section, we derive conditions for the NI state feedback equivalence of the system (1) by investigating an auxiliary system. The auxiliary system is obtained by transforming the system (1) into a form which has a relative degree vector via the use of an output transformation. The existence of such an output transformation will be shown later to be a necessary condition for NI state feedback equivalence. First, we show in the following that the property of NI state feedback equivalence is invariant under a nonsingular output transformation. State feedback equivalence to an NI system is defined as follows.

Definition 8: A system in the form of (1) is said to be state feedback equivalent to an NI system if there exists a state feedback control law

$$u = K_x x + K_v v$$

with $K_x \in \mathbb{R}^{p\times n}$ and $K_v \in \mathbb{R}^{p\times p}$, such that the resulting closed-loop system with the new input $v \in \mathbb{R}^p$ is minimal and NI.

Note that state feedback equivalence problems do not allow for a change of output. However, they allow for a free change of inputs. We show in the following two lemmas that for system (1), its NI state feedback equivalence property is invariant under a nonsingular output transformation.

Lemma 6: Suppose $T \in \mathbb{R}^{p\times p}$ is nonsingular. Then, the transfer matrix $TR(s)T^T$ is NI if and only if $R(s)$ is NI.

Proof: The proof is based on Definition 1. $R(s)$ is NI, if and only if Conditions 1, 2, and 3 in Definition 1 are satisfied. However, the positive definiteness (semidefiniteness) of the matrices in Conditions 1–3 in Definition 1 is invariant under the transformation $R(s) \mapsto TR(s)T^T$. ∎

Lemma 7: Consider the system (1) and the state, input, and output transformations $\tilde{x} = T_x x$, $\tilde{u} = T_u u$, and $\tilde{y} = T_y y$, where $T_x \in \mathbb{R}^{n\times n}$, $T_u, T_y \in \mathbb{R}^{p\times p}$ are nonsingular. Then, the system (1) is state feedback equivalent to an NI system, if and only if the transformed system is also state feedback equivalent to an NI system.

Proof: If the transformed system with state $\tilde{x}$, input $\tilde{u}$, and output $\tilde{y}$ is state feedback equivalent to an NI system, then there exists a control law

$$\tilde{u} = K_x \tilde{x} + K_v \tilde{v}$$

under which the system with input $\tilde{v}$ and output $\tilde{y}$ is minimal and NI. According to Lemma 6, now the system with output $y = T_y^{-1}\tilde{y}$ and input $v = T_y^T \tilde{v}$ is also minimal and NI. This means that the original system with state x, input u, and output y is also state feedback equivalent to an NI system. The corresponding feedback control law can be derived as follows:

$$u = T_u^{-1}\tilde{u} = T_u^{-1}(K_x\tilde{x} + K_v\tilde{v}) = T_u^{-1}(K_x T_x x + K_v T_y^{-T} v). \tag{4}$$

This completes the sufficiency part of the proof. Since the state, input, and output transformation matrices T_x, T_u, and T_y are all nonsingular, the necessity part of the proof follows in the same manner as the sufficiency part with the inverses of the transformations considered. ∎

Consider the case that there exists an output transformation $\tilde{y} = T_y y$, where $T_y \in \mathbb{R}^{p\times p}$ and $\det(T_y) \neq 0$, that transforms the system (1) into a form with a relative degree vector $r = \{r_1, \ldots, r_p\}$. Let $\tilde{\mathcal{C}} = T_y \mathcal{C} \in \mathbb{R}^{p\times n}$, then we can write the transformed system in the form

$$\dot{x} = \mathcal{A}x + \mathcal{B}u, \tag{5a}$$

$$\tilde{y} = \tilde{\mathcal{C}}x \tag{5b}$$

where $rank(\mathcal{B}) = rank(\tilde{\mathcal{C}}) = p$. For ease of presentation of our main NI state feedback equivalence results, we aim to transform the system (5) into a normal form. We show in the following lemma that this is always achievable via input and state transformations in the case that the system (5) has relative degree less than or equal to two.

Lemma 8: Suppose the system (5) has relative degree less than or equal to two. Then, there exist input and state transformations that transform (5) into the following normal form:

$$\dot{z} = A_{00}z + A_{01}x_1 + A_{02}x_2 + A_{03}x_3, \tag{6a}$$

$$\dot{x}_1 = A_{10}z + A_{11}x_1 + A_{12}x_2 + A_{13}x_3 + u_1, \tag{6b}$$

$$\dot{x}_2 = x_3, \tag{6c}$$

$$\dot{x}_3 = A_{30}z + A_{31}x_1 + A_{32}x_2 + A_{33}x_3 + u_2, \tag{6d}$$

$$\tilde{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{6e}$$

where $\tilde{x} = [z^T \quad x_1^T \quad x_2^T \quad x_3^T]^T \in \mathbb{R}^n$ is the state, $\tilde{u} = [u_1^T \quad u_2^T]^T \in \mathbb{R}^p$ is the input, and $\tilde{y} \in \mathbb{R}^p$ is the output of the transformed system. Here, $x_1, u_1 \in \mathbb{R}^{p_1}$ and $x_2, x_3, u_2 \in \mathbb{R}^{p_2}$, where $0 \le p_1 \le p$ and $p_2 := p - p_1$. Also, $z \in \mathbb{R}^m$, where $m := n - p - p_2$.

Proof: Without loss of generality, suppose the components in the relative degree vector r of the system (5) are sorted in nondecreasing order, i.e., $r = \{1, \ldots, 1, 2 \cdots, 2\}$. Let p_1 $(0 \le$

$p_1 \leq p$) be the number of ones in r and $p_2 = p - p_1$ be the number of twos in r. Also, define the following matrices:

$$\tilde{\mathcal{C}}_O = \begin{bmatrix} \tilde{\mathcal{C}}_1 \\ \vdots \\ \tilde{\mathcal{C}}_{p_1} \end{bmatrix} \in \mathbb{R}^{p_1 \times n}, \quad \text{and} \quad \tilde{\mathcal{C}}_T = \begin{bmatrix} \tilde{\mathcal{C}}_{p_1+1} \\ \vdots \\ \tilde{\mathcal{C}}_p \end{bmatrix} \in \mathbb{R}^{p_2 \times n}$$

where $\tilde{\mathcal{C}}_i$ is the ith row in the matrix $\tilde{\mathcal{C}}$. Hence, $\tilde{\mathcal{C}}_O$ is the block matrix in $\tilde{\mathcal{C}}$ which determines the output entries corresponding to the ones in r. $\tilde{\mathcal{C}}_T$ is the block matrix in $\tilde{\mathcal{C}}$ which determines the output entries corresponding to the twos in r. According to Definition 5, we have that $rank(\tilde{\mathcal{C}}_O\mathcal{B}) = p_1$, $\tilde{\mathcal{C}}_T\mathcal{B} = 0$ and $rank(\tilde{\mathcal{C}}_T\mathcal{A}\mathcal{B}) = p_2$. Also, Condition 2 in Definition 5 implies that

$$\det \begin{bmatrix} \tilde{\mathcal{C}}_O\mathcal{B} \\ \tilde{\mathcal{C}}_T\mathcal{A}\mathcal{B} \end{bmatrix} \neq 0. \tag{7}$$

Therefore, the rows of the matrix $\left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]$ are linearly independent. Since $rank(\mathcal{C}) = p$ and $\det T_y \neq 0$, then $rank(\tilde{\mathcal{C}}) = p$. Hence, $\tilde{\mathcal{C}}_T$ has full row rank. Also, according to Condition 1 in Definition 5, we have that $\tilde{\mathcal{C}}_T\mathcal{B} = 0$. Then, we can prove by contradiction that the rows of $\tilde{\mathcal{C}}_T$ are linearly independent of the rows of $\left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]$. Indeed, suppose there exists a row $(\tilde{\mathcal{C}}_T)_\kappa$ of $\tilde{\mathcal{C}}_T$, which is a linear combination of the rows of $\left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]$. Then, $(\tilde{\mathcal{C}}_T)_\kappa\mathcal{B} \neq 0$ according to (7), which contradicts the equation $\tilde{\mathcal{C}}_T\mathcal{B} = 0$. Therefore, the matrix $\left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]$ has full row rank. Define the new state as

$$\begin{aligned} x_1 &= \tilde{\mathcal{C}}_O x, \\ x_2 &= \tilde{\mathcal{C}}_T x, \\ x_3 &= \dot{x}_2 = \tilde{\mathcal{C}}_T\mathcal{A}x. \end{aligned}$$

We also need a complementary state $z \in \mathbb{R}^m$ where $m := n - p - p_2 \geq 0$. Let $z = \tilde{\mathcal{C}}_z x$, where $\tilde{\mathcal{C}}_z$ is such that $\tilde{\mathcal{C}}_z\mathcal{B} = 0$ and the matrix $T_x = \left[\begin{smallmatrix} \tilde{\mathcal{C}}_z \\ \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]$ is nonsingular. Let $\tilde{x} = T_x x$ be the new state. Also, let $\tilde{u} = \left[\begin{smallmatrix} u_1 \\ u_2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]\mathcal{B}u$. According to (7), the input transformation matrix $T_u = \left[\begin{smallmatrix} \tilde{\mathcal{C}}_O \\ \tilde{\mathcal{C}}_T\mathcal{A} \end{smallmatrix}\right]\mathcal{B}$ is nonsingular. The new system has a state-space model

$$\frac{d}{dt}\begin{bmatrix} z \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = T_x\mathcal{A}T_x^{-1}\begin{bmatrix} z \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ u_1 \\ 0 \\ u_2 \end{bmatrix}, \tag{8a}$$

$$\tilde{y} = \begin{bmatrix} 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}\begin{bmatrix} z \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}. \tag{8b}$$

By considering the blocks of $T_x\mathcal{A}T_x^{-1}$ including the relation $\dot{x}_2 = x_3$, we can write (8) in the form (6). ■

We now consider necessary and sufficient conditions under which the system (6) is state feedback equivalent to an NI system. For the system (6), choose the control inputs u_1 and u_2 to be

$$\begin{aligned} u_1 = {} & v_1 + (K_{10} - A_{10})z + (K_{11} - A_{11})x_1 \\ & + (K_{12} - A_{12})x_2 + (K_{13} - A_{13})x_3; \end{aligned} \tag{9}$$

$$\begin{aligned} u_2 = {} & v_2 + (K_{20} - A_{30})z + (K_{21} - A_{31})x_1 \\ & + (K_{22} - A_{32})x_2 + (K_{23} - A_{33})x_3 \end{aligned} \tag{10}$$

which allows the system (6) to be represented by the form

$$\dot{\tilde{x}} = A\tilde{x} + B\tilde{v}, \tag{11a}$$

$$\tilde{y} = C\tilde{x} \tag{11b}$$

where $\tilde{v} = \left[\begin{smallmatrix} v_1 \\ v_2 \end{smallmatrix}\right]$ is the new input and

$$A = \begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} \\ K_{10} & K_{11} & K_{12} & K_{13} \\ 0 & 0 & 0 & I \\ K_{20} & K_{21} & K_{22} & K_{23} \end{bmatrix}, \tag{12}$$

$$B = \begin{bmatrix} 0 & 0 \\ I & 0 \\ 0 & 0 \\ 0 & I \end{bmatrix}, \tag{13}$$

$$C = \begin{bmatrix} 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \end{bmatrix}. \tag{14}$$

We need to find the state feedback matrices

$$\begin{aligned} & K_{10} \in \mathbb{R}^{p_1 \times m}, K_{11} \in \mathbb{R}^{p_1 \times p_1}, K_{12} \in \mathbb{R}^{p_1 \times p_2}, \\ & K_{13} \in \mathbb{R}^{p_1 \times p_2}, K_{20} \in \mathbb{R}^{p_2 \times m}, K_{21} \in \mathbb{R}^{p_2 \times p_1}, \\ & K_{22} \in \mathbb{R}^{p_2 \times p_2}, \text{ and } K_{23} \in \mathbb{R}^{p_2 \times p_2} \end{aligned} \tag{15}$$

such that the system (11) is minimal and NI. The following lemma provides necessary and sufficient conditions for such state feedback matrices to exist.

Lemma 9: Suppose the system (6) satisfies $\det A_{00} \neq 0$. Then, it is state feedback equivalent to an NI system if and only if it is controllable and A_{00} is Lyapunov stable.

Proof: The system (6) is state feedback equivalent to an NI system if and only if there exist state feedback matrices (15) such that the system (11) is NI and the realization (A, B, C) in (12)–(14) is minimal.

First, we prove that the controllability of the system (6) is equivalent to the controllability of the system (11). Let

$$\breve{A} = \begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} \\ A_{10} & A_{11} & A_{12} & A_{13} \\ 0 & 0 & 0 & I \\ A_{30} & A_{31} & A_{32} & A_{33} \end{bmatrix}.$$

Then, we need to prove that the controllability of $(\breve{A}, B)$ is equivalent to that of (A, B). Applying the eigenvector test for controllability (see [42]), $(\breve{A}, B)$ is controllable if and only if any nonzero vector in the kernel of B^T is not an eigenvector of $\breve{A}^T$. Considering the structure of B in (13), a nonzero vector $\eta \in \ker(B^T)$ must take the form $\eta = [\eta_1^T \quad 0 \quad \eta_3^T \quad 0]^T$, where $\eta_1 \neq 0$ or $\eta_3 \neq 0$. Therefore, for any scalar λ_c, we have that

$\breve{A}^T\eta \neq \lambda_c\eta$. Substituting for $\breve{A}$, we obtain

$$\begin{bmatrix} A_{00}^T\eta_1 \\ A_{01}^T\eta_1 \\ A_{02}^T\eta_1 \\ A_{03}^T\eta_1+\eta_3 \end{bmatrix} \neq \lambda_c \begin{bmatrix} \eta_1 \\ 0 \\ \eta_3 \\ 0 \end{bmatrix} \tag{16}$$

for any scalar λ_c. This condition depends only on the matrices A_{00}, A_{01}, A_{02}, and A_{03}, which form the common first block row of the matrices $\breve{A}$ and A. Hence, the controllability of $(\breve{A}, B)$ is equivalent to that of (A, B).

Sufficiency: The condition (16) is satisfied if and only if for any eigenvector η_1 of A_{00}^T with eigenvalue λ_c, we have that $A_{01}^T\eta_1 \neq 0$ or $\begin{bmatrix} A_{02}^T\eta_1 \\ A_{03}^T\eta_1+\eta_3 \end{bmatrix} \neq \lambda_c \begin{bmatrix} \eta_3 \\ 0 \end{bmatrix}$. The condition $A_{01}^T\eta_1 \neq 0$ holds if and only if (A_{00}, A_{01}) is controllable. The condition $\begin{bmatrix} A_{02}^T\eta_1 \\ A_{03}^T\eta_1+\eta_3 \end{bmatrix} \neq \lambda_c \begin{bmatrix} \eta_3 \\ 0 \end{bmatrix}$ holds if and only if for any $\eta_3 = -A_{03}^T\eta_1$, we have that $A_{02}^T\eta_1 \neq \lambda_c\eta_3 = -\lambda_c A_{03}^T\eta_1 = -A_{03}^T A_{00}^T\eta_1$. That is $(A_{03}^T A_{00}^T + A_{02}^T)\eta_1 \neq 0$, which holds if and only if $(A_{00}, A_{00}A_{03} + A_{02})$ is controllable. Therefore, we conclude that (A, B) is controllable if and only if (A_{00}, A_{01}) or $(A_{00}, A_{00}A_{03} + A_{02})$ is controllable.

We now derive necessary and sufficient conditions under which (A, C) is observable. Given the structure of C in (14), any nonzero vector $\sigma \in \ker(C)$ must take the form $\sigma = \begin{bmatrix} \sigma_1^T & 0 & 0 & \sigma_4^T \end{bmatrix}^T$, where $\sigma_1 \neq 0$ or $\sigma_4 \neq 0$. According to the eigenvector test for observability (see [42]), (A, C) is observable if and only if $A\sigma \neq \lambda_o\sigma$ for any scalar λ_o. Substituting A from (12), we obtain

$$\begin{bmatrix} A_{00}\sigma_1 + A_{03}\sigma_4 \\ K_{10}\sigma_1 + K_{13}\sigma_4 \\ \sigma_4 \\ K_{20}\sigma_1 + K_{23}\sigma_4 \end{bmatrix} \neq \lambda_o \begin{bmatrix} \sigma_1 \\ 0 \\ 0 \\ \sigma_4 \end{bmatrix}. \tag{17}$$

When $\sigma_4 \neq 0$, (17) is always true. Now, we consider the case that $\sigma_1 \neq 0$ and $\sigma_4 = 0$. In this case, (17) becomes

$$\begin{bmatrix} A_{00}\sigma_1 \\ K_{10}\sigma_1 \\ 0 \\ K_{20}\sigma_1 \end{bmatrix} \neq \lambda_o \begin{bmatrix} \sigma_1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

which holds if and only if $K_{10}\sigma_1 \neq 0$ or $K_{20}\sigma_1 \neq 0$ for any eigenvector σ_1 of A_{00}. Therefore, we conclude that (A, C) is observable if and only if (A_{00}, K_{10}) or (A_{00}, K_{20}) is observable.

The nonsingular matrix A_{00} is Lyapunov stable if and only if there exists a state transformation $A_{00} \mapsto SA_{00}S^{-1}$ which allows A_{00} to be represented, without loss of generality, as $A_{00} = \mathrm{diag}(A_{00}^a, A_{00}^b)$, where

$$\begin{aligned} &spec(A_{00}^a) \subset j\mathbb{R}\backslash\{0\}, \quad spec(A_{00}^b) \subset \mathbb{C}^-, \\ &\text{and} \quad A_{00}^a + (A_{00}^a)^T = 0. \end{aligned} \tag{18}$$

Here, $A_{00}^a \in \mathbb{R}^{m_a \times m_a}$ and $A_{00}^b \in \mathbb{R}^{m_b \times m_b}$, where $0 \leq m_a \leq m$ and $m_b := m - m_a$. The conditions in (18) are achievable according to the proof in [43, Proposition 11.9.6]. Decomposing $A_{01}, A_{02}, A_{03}, K_{10}$, and K_{20} accordingly using the same state-space transformation, we can write (11) as

$$\dot{z}_1 = A_{00}^a z_1 + A_{01}^a x_1 + A_{02}^a x_2 + A_{03}^a x_3, \tag{19a}$$

$$\dot{z}_2 = A_{00}^b z_2 + A_{01}^b x_1 + A_{02}^b x_2 + A_{03}^b x_3, \tag{19b}$$

$$\dot{x}_1 = K_{10}^a z_1 + K_{10}^b z_2 + K_{11}x_1 + K_{12}x_2 + K_{13}x_3 + v_1, \tag{19c}$$

$$\dot{x}_2 = x_3, \tag{19d}$$

$$\dot{x}_3 = K_{20}^a z_1 + K_{20}^b z_2 + K_{21}x_1 + K_{22}x_2 + K_{23}x_3 + v_2, \tag{19e}$$

$$y = C \begin{bmatrix} z_1 \\ z_2 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}, \; C = \begin{bmatrix} 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & I & 0 \end{bmatrix}. \tag{19f}$$

Since A_{00}^b is Hurwitz, there exist $\mathcal{Y}_1^b = (\mathcal{Y}_1^b)^T > 0$ and $Q_b = Q_b^T > 0$ such that

$$A_{00}^b\mathcal{Y}_1^b + \mathcal{Y}_1^b(A_{00}^b)^T = -Q_b.$$

Let K_{20} be defined as

$$K_{20} = \begin{bmatrix} K_{20}^a & K_{20}^b \end{bmatrix} \tag{20}$$

where

$$\begin{aligned} K_{20}^a &= (A_{02}^a)^T (A_{00}^a)^{-1} - (A_{03}^a)^T; \\ K_{20}^b &= \left(-(A_{02}^b)^T(A_{00}^b)^{-T} - (A_{03}^b)^T + \mathcal{H}\right)(\mathcal{Y}_1^b)^{-1}. \end{aligned} \tag{21}$$

Here, $\mathcal{H}$ is contained in the set

$$S_{\mathcal{H}} = \{\mathcal{H} \in \mathbb{R}^{p_2 \times m_b} | \mathcal{H}^T\mathcal{H} \leq Q_b\}. \tag{22}$$

If $(A_{00}, A_{00}A_{03} + A_{02})$ is controllable, we can always find $\mathcal{H}$ such that (A_{00}, K_{20}) is observable. This is proved as follows. The controllability of $(A_{00}, A_{00}A_{03} + A_{02})$ implies that no eigenvector of $\mathrm{diag}((A_{00}^a)^T, (A_{00}^b)^T)$ is in the kernel of $\begin{bmatrix} (A_{03}^a)^T(A_{00}^a)^T + (A_{02}^a)^T & (A_{03}^b)^T(A_{00}^b)^T + (A_{02}^b)^T \end{bmatrix}$. This implies that both $(A_{00}^a, A_{00}^a A_{03}^a + A_{02}^a)$ and $(A_{00}^b, A_{00}^b A_{03}^b + A_{02}^b)$ are controllable, which can be proved by applying the eigenvector test for controllability to the vectors $\begin{bmatrix} \eta_a \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ \eta_b \end{bmatrix}$, where η_a and η_b are eigenvectors of $(A_{00}^a)^T$ and $(A_{00}^b)^T$, respectively. The pair (A_{00}, K_{20}) is observable if and only if for any nonzero vector $\delta_K = \begin{bmatrix} \delta_a \\ \delta_b \end{bmatrix}$, which is an eigenvector of A_{00}, we have $K_{20}\delta_K \neq 0$. Since A_{00}^a and A_{00}^b have no common eigenvalues, then δ_K is an eigenvector of A_{00} only if $\delta_a = 0$ or $\delta_b = 0$. We consider the following two cases.

1) *Case 1:* $\delta_a \neq 0$ and $\delta_b = 0$. In this case, δ_a is an eigenvector of A_{00}^a; i.e., $A_{00}^a\delta_a = \lambda_a\delta_a$ for some scalar λ_a. Considering (18), δ_a is also an eigenvector of $(A_{00}^a)^T$. Because $(A_{00}^a, A_{00}^a A_{03}^a + A_{02}^a)$ is controllable, $((A_{03}^a)^T(A_{00}^a)^T + (A_{02}^a)^T)\delta_a \neq 0$. Therefore

$$\begin{aligned} K_{20}\delta_K = K_{20}^a\delta_a &= \left((A_{02}^a)^T(A_{00}^a)^{-1} - (A_{03}^a)^T\right)\delta_a \\ &= \left((A_{02}^a)^T - (A_{03}^a)^T A_{00}^a\right)(A_{00}^a)^{-1}\delta_a \end{aligned}$$

$$= \frac{1}{\lambda_a}\left((A_{02}^a)^T + (A_{03}^a)^T (A_{00}^a)^T\right)\delta_a \neq 0$$

where (18) is also used.

2) *Case 2:* $\delta_a = 0$ and $\delta_b \neq 0$. In this case, δ_b is an eigenvector of A_{00}^b. Because $-(A_{02}^b)^T (A_{00}^b)^{-T} - (A_{03}^b)^T$ in (21) is fixed, and $S_{\mathcal{H}}$ has nonempty interior due to the positive definiteness of Q_b, then we can always find $\mathcal{H}$ such that

$$\begin{aligned} K_{20}\delta_K =&\ K_{20}^b \delta_b \\ =&\ \Big(-(A_{02}^b)^T (A_{00}^b)^{-T} - (A_{03}^b)^T \\ &+ \mathcal{H}\Big) (\mathcal{Y}_1^b)^{-1}\delta_b \neq 0 \end{aligned}$$

for all δ_b that are eigenvalues of A_{00}^b. We conclude that there exists $\mathcal{H}$ such that (A_{00}, K_{20}) is observable.

We choose such matrix $\mathcal{H}$ in the following proof. Let K_{10} be defined as

$$K_{10} = \begin{bmatrix} K_{10}^a & K_{10}^b \end{bmatrix} \tag{23}$$

where

$$\begin{aligned} K_{10}^a =&\ (A_{01}^a)^T (A_{00}^a)^{-1}; \\ K_{10}^b =&\ -\left((A_{01}^b)^T (A_{00}^b)^{-T} + K_{13}\mathcal{H}\right)(\mathcal{Y}_1^b)^{-1}. \end{aligned}$$

Here, K_{13} is contained in the set

$$S_K = \{K_{13} \in \mathbb{R}^{p_1\times p_2} | K_{13}K_{13}^T \leq 2I\}. \tag{24}$$

If (A_{00}, A_{01}) is controllable, we can always find K_{13} such that (A_{00}, K_{10}) is observable. This is proved as follows. The controllability of (A_{00}, A_{01}) implies that no eigenvector of $\mathrm{diag}((A_{00}^a)^T, (A_{00}^b)^T)$ is in $ker(\begin{bmatrix}(A_{01}^a)^T & (A_{01}^b)^T\end{bmatrix})$. This implies that both (A_{00}^a, A_{01}^a) and (A_{00}^b, A_{01}^b) are controllable, which can be proved by applying the eigenvector test for controllability to the vectors $\begin{bmatrix}\eta_a \\ 0\end{bmatrix}$ and $\begin{bmatrix}0 \\ \eta_b\end{bmatrix}$, where η_a and η_b are eigenvectors of $(A_{00}^a)^T$ and $(A_{00}^b)^T$, respectively. The pair (A_{00}, K_{10}) is observable if and only if for any nonzero vector $\phi_K = \begin{bmatrix}\phi_a \\ \phi_b\end{bmatrix}$, which is an eigenvector of A_{00}, we have that $K_{10}\phi_K \neq 0$. Since A_{00}^a and A_{00}^b have no common eigenvalues, then ϕ_K is an eigenvector of A_{00} only if $\phi_a = 0$ or $\phi_b = 0$. We consider the following two cases.

1) *Case 1:* $\phi_a \neq 0$ and $\phi_b = 0$. In this case, ϕ_a is an eigenvector of A_{00}^a; i.e., $A_{00}^a\phi_a = \mu_a\phi_a$ for some scalar μ_a. Considering (18), ϕ_a is also an eigenvector of $(A_{00}^a)^T$. Because (A_{00}^a, A_{01}^a) is controllable, $(A_{01}^a)^T\phi_a \neq 0$. Therefore

$$\begin{aligned} K_{10}\phi_K = K_{10}^a\phi_a = (A_{01}^a)^T (A_{00}^a)^{-1}\phi_a \\ = \frac{1}{\mu_a}(A_{01}^a)^T\phi_a \neq 0 \end{aligned}$$

where (18) is also used.

2) *Case 2:* $\phi_a = 0$ and $\phi_b \neq 0$. In this case, ϕ_b is an eigenvector of A_{00}^b. Because $(A_{01}^b)^T (A_{00}^b)^{-T}$ is fixed and the set S_K has a nonempty interior, then we can always find K_{13}, together with an $\mathcal{H}$ that makes (A_{00}, K_{20}) observable, such that

$$\begin{aligned} K_{10}\phi_K =&\ K_{10}^b\phi_b \\ =&\ -\left((A_{01}^b)^T \left(A_{00}^b\right)^{-T} + K_{13}\mathcal{H}\right)(\mathcal{Y}_1^b)^{-1}\phi_b \neq 0 \end{aligned}$$

for all ϕ_b that are eigenvectors of A_{00}^b. Therefore, with this particular choice of K_{13}, we have that (A_{00}, K_{10}) is observable.

Now recall that (A, B) is controllable if and only if (A_{00}, A_{01}) or $(A_{00}, A_{00}A_{03} + A_{02})$ is controllable. Also, (A, C) is observable if and only if (A_{00}, K_{10}) or (A_{00}, K_{20}) is observable. Since the controllability of (A_{00}, A_{01}) implies the observability of (A_{00}, K_{10}) and the controllability of $(A_{00}, A_{00}A_{03} + A_{02})$ implies the observability of (A_{00}, K_{20}), then the controllability of (A, B) implies the observability of (A, C), when suitable $\mathcal{H}$ and K_{13} are chosen. Therefore, with those choices of $\mathcal{H}$ and K_{13}, the controllability of the system (6) implies that the realization (A, B, C) in (12), (13), and (14) is minimal. Choose the other state feedback matrices as follows:

$$K_{11} = K_{10}A_{00}^{-1}A_{01} - \mathcal{Y}_2^{-1}, \tag{25}$$

$$K_{12} = K_{10}A_{00}^{-1}A_{02}, \tag{26}$$

$$K_{21} = K_{20}A_{00}^{-1}A_{01}, \tag{27}$$

$$K_{22} = K_{20}A_{00}^{-1}A_{02} - \mathcal{Y}_3^{-1}, \tag{28}$$

$$K_{23} = -\frac{1}{2}I \tag{29}$$

where $\mathcal{Y}_2 \in \mathbb{R}^{p_1\times p_1}$ and $\mathcal{Y}_3 \in \mathbb{R}^{p_2\times p_2}$ can be any symmetric positive definite matrices; i.e., $\mathcal{Y}_2 = \mathcal{Y}_2^T > 0$ and $\mathcal{Y}_3 = \mathcal{Y}_3^T > 0$. We will apply Lemma 1 in the following in order to prove that the system (11) is an NI system. We construct the matrix Y as follows:

$$Y = \begin{bmatrix} Y_{11} & -A_{00}^{-1}A_{01}\mathcal{Y}_2 & -A_{00}^{-1}A_{02}\mathcal{Y}_3 & 0 \\ -\mathcal{Y}_2A_{01}^TA_{00}^{-T} & \mathcal{Y}_2 & 0 & 0 \\ -\mathcal{Y}_3A_{02}^TA_{00}^{-T} & 0 & \mathcal{Y}_3 & 0 \\ 0 & 0 & 0 & I \end{bmatrix} \tag{30}$$

where $Y_{11} = \mathcal{Y}_1 + A_{00}^{-1}A_{01}\mathcal{Y}_2A_{01}^TA_{00}^{-T} + A_{00}^{-1}A_{02}\mathcal{Y}_3A_{02}^TA_{00}^{-T}$. Here, $\mathcal{Y}_1 = \mathrm{diag}(y_1^aI, \mathcal{Y}_1^b)$ with $y_1^a > 0$ being a scalar. It can be verified that $Y > 0$ using the Schur complement theorem.

In order to verify Condition 1 in Lemma 1, we note that for the determinant of the matrix A in (12) we have

$$\begin{aligned} (-1)^{p_2}\det A =&\ \det\begin{bmatrix} A_{00} & A_{01} & A_{02} & A_{03} \\ K_{10} & K_{11} & K_{12} & K_{13} \\ K_{20} & K_{21} & K_{22} & K_{23} \\ 0 & 0 & 0 & I \end{bmatrix} \\ =&\ \det\begin{bmatrix} A_{00} & A_{01} & A_{02} \\ K_{10} & K_{11} & K_{12} \\ K_{20} & K_{21} & K_{22} \end{bmatrix} \\ =&\ \det A_{00}\det\left(\begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} - \begin{bmatrix} K_{10} \\ K_{20} \end{bmatrix} A_{00}^{-1}\begin{bmatrix} A_{01} & A_{02} \end{bmatrix}\right) \end{aligned}$$

$$= \det A_{00} \det \begin{bmatrix} -\mathcal{Y}_2^{-1} & 0 \\ 0 & -\mathcal{Y}_3^{-1} \end{bmatrix}$$
$$= \det A_{00} \det(-\mathcal{Y}_2^{-1}) \det(-\mathcal{Y}_3^{-1}) \neq 0$$

where the equalities also use (25)–(28). Also, the input feedthrough matrix in the system (11) is zero, and hence symmetric. Hence, Condition 1 in Lemma 1 is satisfied. For Condition 2 in Lemma 1, with Y defined in (30), we have

$$AY = \begin{bmatrix} y_1^a A_{00}^a & 0 & 0 & 0 & A_{03}^a \\ 0 & A_{00}^b \mathcal{Y}_1^b & 0 & 0 & A_{03}^b \\ 0 & -K_{13}\mathcal{H} & -I & 0 & K_{13} \\ 0 & 0 & 0 & 0 & I \\ -(A_{03}^a)^T & \mathcal{H} - (A_{03}^b)^T & 0 & -I & -\frac{1}{2}I \end{bmatrix}. \quad (31)$$

Therefore, we have that

$$AYC^T = -B;$$

$$AY + YA^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -\mathcal{Q}_b & -\mathcal{H}^T K_{13}^T & 0 & \mathcal{H}^T \\ 0 & -K_{13}\mathcal{H} & -2I & 0 & K_{13} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \mathcal{H} & K_{13}^T & 0 & -I \end{bmatrix}. \quad (32)$$

Let $M = \begin{bmatrix} -\mathcal{Q}_b & -\mathcal{H}^T K_{13}^T & \mathcal{H}^T \\ -K_{13}\mathcal{H} & -2I & K_{13} \\ \mathcal{H} & K_{13}^T & -I \end{bmatrix}$, where (29) is used. For the matrix $-M$, we have that $I > 0$ and the Schur complement of the block I is

$$(-M)/I = \begin{bmatrix} \mathcal{Q}_b & \mathcal{H}^T K_{13}^T \\ K_{13}\mathcal{H} & 2I \end{bmatrix} - \begin{bmatrix} \mathcal{H}^T \\ K_{13} \end{bmatrix} \begin{bmatrix} \mathcal{H} & K_{13}^T \end{bmatrix}$$
$$= \begin{bmatrix} \mathcal{Q}_b - \mathcal{H}^T\mathcal{H} & 0 \\ 0 & 2I - K_{13}K_{13}^T \end{bmatrix} \geq 0$$

where (22) and (24) are also used. Therefore, we have that $M \leq 0$. Hence, $AY + YA^T \leq 0$, and Condition 2 in Lemma 1 is satisfied. Hence, the system (11) is an NI system with a minimal realization.

Necessity: If the realization (A, B, C) in (12)–(14) is minimal and NI, then according to the proof of Lemma 1 (see in [3, Lemma 7]), there exists an $X = X^T > 0$ such that

$$\begin{bmatrix} XA + A^TX & XB - A^TC^T \\ B^TX - CA & -(CB + B^TC^T) \end{bmatrix} \leq 0.$$

Therefore, for any z, x_1, x_2, x_3, and v, we have

$$\begin{bmatrix} z \\ x_1 \\ x_2 \\ x_3 \\ v \end{bmatrix}^T \begin{bmatrix} XA + A^TX & XB - A^TC^T \\ B^TX - CA & -(CB + B^TC^T) \end{bmatrix} \begin{bmatrix} z \\ x_1 \\ x_2 \\ x_3 \\ v \end{bmatrix} \leq 0. \quad (33)$$

Let $X = \begin{bmatrix} X_{11} & X_{12} & X_{13} & X_{14} \\ X_{12}^T & X_{22} & X_{23} & X_{24} \\ X_{13}^T & X_{23}^T & X_{33} & X_{34} \\ X_{14}^T & X_{24}^T & X_{34}^T & X_{44} \end{bmatrix}$ and substitute (12)–(14) into (33). Also, let $x_1 = 0$, $x_2 = x_3 = 0$, and $\tilde{v} = \begin{bmatrix} -K_{10} \\ -K_{20} \end{bmatrix} z$. We get

$$z^T(X_{11}A_{00} + A_{00}^T X_{11})z \leq 0$$

for any z, which implies that $X_{11}A_{00} + A_{00}^T X_{11} \leq 0$. Considering $X = X^T > 0$, we have $X_{11} > 0$. Also, since $\det A_{00} \neq 0$, then A_{00} is Lyapunov stable. ■

To facilitate the description of the conditions for NI state feedback equivalence to a system in the general form (1), we recall the following terminologies (see [34] and [44]). In the case when the system (5) has relative degree less than or equal to two, the system (6) is said to be the *normal form* of (5). The dynamics (6a), which are not controlled by the input u directly or through chains of integrators, are called the *internal dynamics*. The other part of the state, described by (6b)–(6d), are called the *external dynamics*. Setting the states described by the external dynamics to be zero in the internal dynamics, we obtain the *zero dynamics*

$$\dot{z} = A_{00}z. \quad (34)$$

We now provide the definition of the weakly minimum phase property.

Definition 9: (Weakly minimum phase) [28], [29]. The system (5) of relative degree less than or equal to two is said to be weakly minimum phase if its zero dynamics (34) are Lyapunov stable.

Theorem 1: Suppose the system (1) satisfying $rank(\mathcal{B}) = rank(\mathcal{C}) = p$ is minimal with no zero at the origin. Then, it is state feedback equivalent to an NI system if and only if there exists an output transformation $\tilde{y} = T_y y$, where $T_y \in \mathbb{R}^{p\times p}$ and $\det T_y \neq 0$, such that the transformed system has relative degree less than or equal to two, and the transformed system is weakly minimum phase.

Proof: *Sufficiency.* The sufficiency part of the proof directly follows from Lemmas 7, 8, and 9. According to Lemma 8, the system (1) can always be transformed into the form (6) using nonsingular input, output, and state transformations. Since the system (1) has no zero at the origin, then $\det A_{00} \neq 0$ because nonsingular input, output, and state transformations do not change the zeros of a system. Also, since the transformed system is weakly minimum phase, then A_{00} is Lyapunov stable. Since the input, output, and state transformations are all nonsingular, the minimality of the system (1) is preserved in (6). According to Lemma 9, the output transformed system (6) is state feedback equivalent to an NI system. According to Lemma 7, the original system (1) is also state feedback equivalent to an NI system. This completes the sufficiency part of the proof.

Necessity. We first prove that if the system (1) is state feedback equivalent to an NI system, then there exists an output transformation $\tilde{y} = T_y y$ that transforms the system (1) into a system with relative degree less than or equal to two.

If the system (1) is state feedback equivalent to an NI system, then according to Lemma 6, it is still feedback equivalent NI after a nonsingular output transformation. We apply an output transformation to the system $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ in order that the transformed system has a leading incomplete relative degree vector. Since the output transformed system is feedback equivalent to an NI system, then under a state feedback control law, we can make it NI with a minimal realization $(\widehat{A}, \widehat{B}, \widehat{C})$.

Since the system with realization $(\widehat{A}, \widehat{B}, \widehat{C})$ has a leading incomplete relative degree vector r, we denote by $p_1 \geq 0$ the number of components in r that equal to one; i.e., $r_1, \ldots, r_{p_1} = 1$, and $r_{p_1+1}, \ldots, r_p \geq 2$. We decompose the matrix $\widehat{C}$ as $\widehat{C} = \begin{bmatrix} \widehat{C}_O \\ \widehat{C}_G \end{bmatrix}$, where $\widehat{C}_O \in \mathbb{R}^{p_1 \times n}$ and $\widehat{C}_G \in \mathbb{R}^{(p-p_1)\times n}$. Here, $\widehat{C}_O$ determines the output entries corresponding to the ones in r, and $\widehat{C}_G$ determines the output entries corresponding to the components greater than one in r. According to Definition 6, $rank(\widehat{C}_O \widehat{B}) = p_1$ and $\widehat{C}_G \widehat{B} = 0$.

According to the proof of Lemma 1 (see [3]), the fact that $(\widehat{A}, \widehat{B}, \widehat{C})$ is NI implies that there exists $X = X^T > 0$ such that

$$\begin{bmatrix} X\widehat{A} + \widehat{A}^T X & X\widehat{B} - \widehat{A}^T \widehat{C}^T \\ \widehat{B}^T X - \widehat{C}\widehat{A} & -(\widehat{C}\widehat{B} + \widehat{B}^T \widehat{C}^T) \end{bmatrix} \leq 0. \tag{35}$$

Decomposing $\widehat{B}$ accordingly as $\widehat{B} = [\widehat{B}_O \quad \widehat{B}_G]$ where $\widehat{B}_O \in \mathbb{R}^{n\times p_1}$ and $\widehat{B}_G \in \mathbb{R}^{n\times(p-p_1)}$, the inequality (35) can be expanded to be

$$\begin{bmatrix} X\widehat{A} + \widehat{A}^T X & X\widehat{B}_O - \widehat{A}^T \widehat{C}_O^T & X\widehat{B}_G - \widehat{A}^T \widehat{C}_G^T \\ \widehat{B}_O^T X - \widehat{C}_O \widehat{A} & -(\widehat{C}_O \widehat{B}_O + \widehat{B}_O^T \widehat{C}_O^T) & -\widehat{C}_O \widehat{B}_G \\ \widehat{B}_G^T X - \widehat{C}_G \widehat{A} & -\widehat{B}_G^T \widehat{C}_O^T & 0 \end{bmatrix} \leq 0 \tag{36}$$

where the condition $\widehat{C}_G \widehat{B} = 0$ is also used. The condition (36) implies that $\widehat{C}_O \widehat{B}_G = 0$ and $\widehat{B}_G^T X - \widehat{C}_G \widehat{A} = 0$. We have that $rank(\widehat{B}_G) = p - p_1$ because $rank(\widehat{B}) = rank(\mathcal{B}) = p$. Then, $\widehat{B}_G^T X - \widehat{C}_G \widehat{A} = 0$ implies that $\widehat{C}_G \widehat{A} \widehat{B}_G = \widehat{B}_G^T X \widehat{B}_G > 0$. The positive definiteness of $\widehat{C}_G \widehat{A} \widehat{B}_G$ implies that the largest component in the leading incomplete relative degree vector r of the system is two. Moreover, we have that

$$\begin{bmatrix} \widehat{C}_O \widehat{B} \\ \widehat{C}_G \widehat{A} \widehat{B} \end{bmatrix} = \begin{bmatrix} \widehat{C}_O \widehat{B}_O & 0 \\ \widehat{C}_G \widehat{A} \widehat{B}_O & \widehat{C}_G \widehat{A} \widehat{B}_G \end{bmatrix}. \tag{37}$$

Since $\widehat{C}_O \widehat{B}_G = 0$ and $rank(\widehat{C}_O \widehat{B}) = p_1$, we have that $\det(\widehat{C}_O \widehat{B}_O) \neq 0$. Considering that $\widehat{C}_O \widehat{B}_O$ and $\widehat{C}_G \widehat{A} \widehat{B}_G$ in (37) are both nonsingular, we have that $\det \begin{bmatrix} \widehat{C}_O \widehat{B} \\ \widehat{C}_G \widehat{A} \widehat{B} \end{bmatrix} \neq 0$. This implies that the leading incomplete relative degree vector r of the realization $(\widehat{A}, \widehat{B}, \widehat{C})$ is indeed a relative degree vector, whose components are either one or two. Therefore, we conclude that the system (1) can be output transformed into a system with a relative degree vector $r = \{r_1, \ldots, r_p\}$ with $1 \leq r_i \leq 2$ for all $i = 1, \ldots, p$. Therefore, the system (1) can be transformed into the form (6) using input, output, and state transformations. The necessity part of Lemma 9 implies that the weakly minimum phase property of the output transformed system is another necessary condition. This completes the necessity part of the proof. ■

The proof of the above theorem leads to the Procedure 1 for constructing the control input to make the system (1) NI.

Procedure 1: Construction of state feedback matrices to make the system (1) NI

Step 1. Apply a nonsingular output transformation $\tilde{y} = T_y y$ to transform the system (1) into the system (5), which has relative degree less than or equal to two using the method described in Lemma 4 of [41].

Step 2. Apply nonsingular input and state transformations $\tilde{x} = T_x x$ and $\tilde{u} = T_x u$ such that the resulting system has the normal form (6). (see proof of Lemma 8)

Step 3. Construct the state feedback control inputs for the system (6) using the formulas (9) and (10).

3a). The state feedback matrices K_{20}, K_{10} and K_{13} can be constructed according to (20), (23) and (24) respectively, such that the pairs (A_{00}, K_{20}) and (A_{00}, K_{10}) are both observable.

3b). The state feedback matrices K_{11}, K_{12}, K_{21}, K_{22}, K_{23} can be constructed accordingly using the formulas (25)–(29).

Step 4. Given the state feedback control input obtained in Step 3 for the normal form (6), we calculate the state feedback control input for the original system (1) using (4).

B. State Feedback Equivalence to OSNI Systems

We also derive necessary and sufficient conditions under which the system (1) can be rendered OSNI.

Definition 10: A system in the form of (1) is said to be state feedback equivalent to an OSNI system if there exists a state feedback control law

$$u = K_x x + K_v v$$

where $K_x \in \mathbb{R}^{p\times n}$ and $K_v \in \mathbb{R}^{p\times p}$, such that the closed-loop system with the new input $v \in \mathbb{R}^p$ is minimal and OSNI.

Similar to Lemma 6, we show in the following that the OSNI state feedback equivalence property is invariant under a nonsingular output transformation.

Lemma 10: If the transfer matrix $R(s)$ is OSNI, then $TR(s)T^T$ is also OSNI, where $T \in \mathbb{R}^{p\times p}$ and $\det T \neq 0$.

Proof: The proof follows from Definition 3. If $R(s)$ is OSNI, then we have that

$$j\omega[R(j\omega) - R(j\omega)^*] - \epsilon\omega^2 \bar{R}(j\omega)^* \bar{R}(j\omega) \geq 0$$

$\forall \omega \in \mathbb{R} \cup \{\infty\}$ where $\bar{R}(j\omega) = R(j\omega) - R(\infty)$. We have that $T^T T \leq \lambda_{\max}(T^T T) I$. Therefore,

$$\begin{aligned} & j\omega[R(j\omega) - (R(j\omega))^*] - \frac{\epsilon}{\lambda_{\max}(T^T T)}\omega^2 \bar{R}(j\omega)^* T^T T \bar{R}(j\omega) \\ & \geq j\omega[R(j\omega) - (R(j\omega))^*] - \frac{\epsilon}{\lambda_{\max}(T^T T)}\omega^2 \bar{R}(j\omega)^* T^T T \bar{R}(j\omega) \\ & \quad + \frac{\epsilon}{\lambda_{\max}(T^T T)}\omega^2 \bar{R}(j\omega)^* (T^T T - \lambda_{\max}(T^T T) I) \bar{R}(j\omega) \\ & = j\omega[R(j\omega) - R(j\omega)^*] - \epsilon\omega^2 \bar{R}(j\omega)^* \bar{R}(j\omega) \geq 0. \end{aligned} \tag{38}$$

The transfer matrix $TR(s)T^T$ satisfies Definition 3 via (38). Therefore, the transformed system $TR(s)T^T$ is OSNI with the output strictness $\frac{\epsilon}{\lambda_{\max}(T^TT)}$. ■

We show in the following that the same conditions in Theorem 1 also lead to state feedback equivalence to an OSNI system.

Lemma 11: Suppose the system (6) has $\det A_{00} \neq 0$. Then it is state feedback equivalent to an OSNI system if and only if it is controllable and A_{00} is Lyapunov stable.

Proof: The necessity part of this lemma follows from the necessity part of Lemma 9 because OSNI systems belong to the class of NI systems.

For the sufficiency part, we need to show that the condition

$$AY + YA^T + \epsilon(CAY)^T CAY \leq 0$$

in Lemma 3 is satisfied for some scalar $\epsilon > 0$ in addition to what is shown in the sufficiency proof of Lemma 9. Following from the sufficiency proof of Lemma 9, we add restrictions on the choices of K_{13} and $\mathcal{H}$ such that $K_{13}^T K_{13} = I$ and $\mathcal{H} \in \tilde{S}_{\mathcal{H}}$, where

$$\tilde{S}_{\mathcal{H}} = \left\{ \mathcal{H} \in \mathbb{R}^{p_2 \times m_b} \,\middle|\, \mathcal{H}^T\mathcal{H} \leq \frac{\epsilon^2 - 3\epsilon + 1}{1 - 2\epsilon} Q_b \right\}.$$

Here, we have that $\frac{\epsilon^2-3\epsilon+1}{1-2\epsilon} \in (0, 1)$ for $\epsilon \in (0, \frac{1}{2}(3 - \sqrt{5}))$. Note that this additional restriction does not change the results in Lemma 9. Using C and AY in (19f) and (31), we have that

$$CAY = \begin{bmatrix} 0 & -K_{13}\mathcal{H} & -I & 0 & K_{13} \\ 0 & 0 & 0 & 0 & I \end{bmatrix}.$$

Therefore

$$(CAY)^T CAY = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \mathcal{H}^T\mathcal{H} & \mathcal{H}^T K_{13}^T & 0 & -\mathcal{H}^T \\ 0 & K_{13}\mathcal{H} & I & 0 & -K_{13} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -\mathcal{H} & -K_{13}^T & 0 & 2I \end{bmatrix}.$$

Hence

$$\begin{aligned} &AY + YA^T + \epsilon(CAY)^T CAY \\ &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -Q_b + \epsilon\mathcal{H}^T\mathcal{H} & -(1-\epsilon)\mathcal{H}^T K_{13}^T & 0 & (1-\epsilon)\mathcal{H}^T \\ 0 & -(1-\epsilon)K_{13}\mathcal{H} & -(2-\epsilon)I & 0 & (1-\epsilon)K_{13} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & (1-\epsilon)\mathcal{H} & (1-\epsilon)K_{13}^T & 0 & -(1-2\epsilon)I \end{bmatrix}. \end{aligned}$$

Let

$$\tilde{M} = \begin{bmatrix} -Q_b + \epsilon\mathcal{H}^T\mathcal{H} & -(1-\epsilon)\mathcal{H}^T K_{13}^T & (1-\epsilon)\mathcal{H}^T \\ -(1-\epsilon)K_{13}\mathcal{H} & -(2-\epsilon)I & (1-\epsilon)K_{13} \\ (1-\epsilon)\mathcal{H} & (1-\epsilon)K_{13}^T & -(1-2\epsilon)I \end{bmatrix}.$$

We apply the Schur complement theorem in the following to find the range of ϵ. We choose $\epsilon \in (0, \frac{1}{2})$ and therefore, $-(1 - 2\epsilon)I < 0$. The Schur complement of the block $(1 - 2\epsilon)I$ of $-\tilde{M}$ is

$$\begin{aligned} &(-\tilde{M})/[(1-2\epsilon)I] \\ &= \begin{bmatrix} Q_b - \epsilon\mathcal{H}^T\mathcal{H} & (1-\epsilon)\mathcal{H}^T K_{13}^T \\ (1-\epsilon)K_{13}\mathcal{H} & (2-\epsilon)I \end{bmatrix} \\ &\quad - \frac{(1-\epsilon)^2}{1-2\epsilon} \begin{bmatrix} \mathcal{H}^T \\ K_{13} \end{bmatrix} \begin{bmatrix} \mathcal{H} & K_{13}^T \end{bmatrix} \\ &= \begin{bmatrix} Q_b - \left(\epsilon + \frac{(1-\epsilon)^2}{1-2\epsilon}\right)\mathcal{H}^T\mathcal{H} & \left(1 - \epsilon - \frac{(1-\epsilon)^2}{1-2\epsilon}\right)\mathcal{H}^T K_{13}^T \\ \left(1 - \epsilon - \frac{(1-\epsilon)^2}{1-2\epsilon}\right)K_{13}\mathcal{H} & \left(2 - \epsilon - \frac{(1-\epsilon)^2}{1-2\epsilon}\right)I \end{bmatrix} \end{aligned}$$

which is positive semidefinite when $\epsilon \in (0, \frac{1}{2}(3 - \sqrt{5}))$. In this case, $AY + YA^T + \epsilon(CAY)^T CAY \leq 0$. Therefore, the system with the realization (A, B, C) in (12), (13), and (14) is OSNI. ■

Theorem 2: Suppose the system (1) is minimal with no zero at the origin. Then, it is state feedback equivalent to an OSNI system if and only if there exists an output transformation $\tilde{y} = T_y y$, where $T_y \in \mathbb{R}^{p\times p}$ and $\det T_y \neq 0$, such that the transformed system has relative degree less than or equal to two, and the transformed system is weakly minimum phase.

Proof: This proof is similar to the proof of Theorem 1 except that Lemmas 10 and 11 are used instead of Lemmas 6 and 9. ■

Considering the results in Theorem 1 and 2, we have the following corollary which shows that the same necessary and sufficient conditions are shared for NI and OSNI state feedback equivalence.

Corollary 1: Suppose the system (1) is minimal with no zero at the origin. Then, the following statements are equivalent.

1) The system (1) is state feedback equivalent to an NI system.
2) The system (1) is state feedback equivalent to an OSNI system.
3) There exists an output transformation $\tilde{y} = T_y y$, where $T_y \in \mathbb{R}^{p\times p}$ and $\det T_y \neq 0$, such that the transformed system has relative degree less than or equal to two, and the transformed system is weakly minimum phase.

Remark 2: Although the necessary and sufficient conditions for state feedback equivalence to NI and OSNI systems are the same, as shown in Corollary 1, different state feedback matrices are required. That is, additional restrictions on state feedback matrices should be satisfied in order to make a system OSNI. Considering that under certain assumptions, the interconnection of an NI system and an OSNI system is asymptotically stable [8], [26], we can decide to make a system NI or OSNI depending on the other system in the interconnection.

IV. State Feedback Equivalence to an SSNI System

In this section, we derive necessary and sufficient conditions under which a system in the form of (1) is state feedback equivalent to an SSNI system. First, we define state feedback equivalence to an SSNI system as follows.

Definition 11: A system in the form of (1) is said to be state feedback equivalent to an SSNI system if there exists a state feedback control law

$$u = K_x x + K_v v$$

where $K_x \in \mathbb{R}^{p\times n}$ and $K_v \in \mathbb{R}^{p\times p}$, such that the closed-loop system with the new input $v \in \mathbb{R}^p$ is SSNI.

It will be shown later in this section that having a relative degree vector $r = \{1, \ldots, 1\}$ is one of the necessary conditions for this system to be state feedback equivalent to an SSNI system. Therefore, we start with the derivation of the normal form for the system (1) with a relative degree vector $r = \{1, \dots, 1\}$.

Lemma 12: Suppose the system (1) satisfying $rank(\mathcal{B}) = rank(\mathcal{C}) = p$ has a relative degree vector $r = \{1, \dots, 1\}$. Then, there exists input and state transformations such that the resulting transformed system is of the form

$$\dot{z} = A_{00}z + A_{01}y, \tag{39a}$$

$$\dot{x}_1 = A_{10}z + A_{11}x_1 + \tilde{u}, \tag{39b}$$

$$y = \begin{bmatrix} 0 & I \end{bmatrix} \begin{bmatrix} z \\ x_1 \end{bmatrix}. \tag{39c}$$

Proof: If (1) has a relative degree vector $r = \{1, \dots, 1\}$, then $\det(\mathcal{CB}) \neq 0$. The rest of the proof follows from Lemma 8 with $p_1 = p$ and $p_2 = 0$. ∎

Choose the input u to be

$$\tilde{u} = v + (K_1 - A_{10})z + (K_2 - A_{11})y$$

and the system (39) takes the form

$$\dot{z} = A_{00}z + A_{01}y, \tag{40a}$$

$$\dot{y} = K_1 z + K_2 y + v, \tag{40b}$$

$$y = \begin{bmatrix} 0 & I \end{bmatrix} \begin{bmatrix} z \\ y \end{bmatrix}. \tag{40c}$$

We need to find the state feedback matrices $K_1 \in \mathbb{R}^{p\times m}$ and $K_2 \in \mathbb{R}^{p\times p}$ such that the system (40) is SSNI.

Lemma 13: [35] Suppose the system (40) has (A_{00}, A_{01}) controllable. Then, the following statements are equivalent.

1. A_{00} is Hurwitz.
2. There exist K_1 and K_2 such that the system (40) is an SSNI system with realization (A, B, C), where A is Hurwitz, and the transfer function $R(s) := C(sI - A)^{-1}B$ is such that $R(s) + R(-s)^T$ has full normal rank.

Proof: Let us define the following:

$$A = \begin{bmatrix} A_{00} & A_{01} \\ K_1 & K_2 \end{bmatrix}, \tag{41}$$

$$B = \begin{bmatrix} 0 \\ I \end{bmatrix}, \tag{42}$$

$$C = \begin{bmatrix} 0 & I \end{bmatrix}. \tag{43}$$

From the proof of Lemma 9, (A, B) is controllable if and only if (A_{00}, A_{01}) is controllable. Therefore, there are no observable uncontrollable modes in this system.

Sufficiency: Let A_{00} be Hurwitz. Then, we can always find a matrix $\mathcal{Y}_1 > 0$ such that

$$A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T + \frac{1}{2}A_{00}^{-1}A_{01}A_{01}^T A_{00}^{-T} < 0$$

is satisfied. In the sequel, we will find a matrix K_1 such that

$$A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T + \frac{1}{2}(\mathcal{Y}_1 K_1^T + A_{00}^{-1}A_{01})(K_1\mathcal{Y}_1 + A_{01}^T A_{00}^{-T}) < 0 \tag{44}$$

is satisfied. One possible choice is $K_1 = -A_{01}^T A_{00}^{-T}\mathcal{Y}_1^{-1}$, which simplifies (44) to be $A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T < 0$. Let $K_2 = K_1 A_{00}^{-1}A_{01} - \mathcal{Y}_2^{-1}$, where $\mathcal{Y}_2 \in \mathbb{R}^{p\times p}$ can be any symmetric positive definite matrix; i.e., $\mathcal{Y}_2 = \mathcal{Y}_2^T > 0$. We apply Lemma 2 in the following to prove that the system (40) is an SSNI system. We construct the matrix Y as follows:

$$Y = \begin{bmatrix} \mathcal{Y}_1 + A_{00}^{-1}A_{01}\mathcal{Y}_2 A_{01}^T A_{00}^{-T} & -A_{00}^{-1}A_{01}\mathcal{Y}_2 \\ -\mathcal{Y}_2 A_{01}^T A_{00}^{-T} & \mathcal{Y}_2 \end{bmatrix}.$$

We have $Y > 0$ because $\mathcal{Y}_2 > 0$ and the Schur complement of the block $\mathcal{Y}_2$ is $\mathcal{Y}_1$, which is positive definite. Now, we have $B + AYC^T = 0$ and

$$AY + YA^T = \begin{bmatrix} A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T & \mathcal{Y}_1 K_1^T + A_{00}^{-1}A_{01} \\ K_1\mathcal{Y}_1 + A_{01}^T A_{00}^{-T} & -2I \end{bmatrix}.$$

We have $-2I < 0$ and the Schur complement of the block $2I$ in the matrix $-(AY + YA^T)$ is

$$\begin{aligned} &(-(AY + YA^T))/(2I) \\ &= -A_{00}\mathcal{Y}_1 - \mathcal{Y}_1 A_{00}^T \\ &\quad - \frac{1}{2}(\mathcal{Y}_1 K_1^T + A_{00}^{-1}A_{01})(K_1\mathcal{Y}_1 + A_{01}^T A_{00}^{-T}) > 0 \end{aligned}$$

according to (44). Hence, $AY + YA^T < 0$ and A^T is Hurwitz. Therefore A is Hurwitz. Now we prove that $R(s) + R(-s)^T$ has full normal rank. For A, B, and C given by (41)–(43), we have

$$\begin{aligned} R(s) &= C(sI - A)^{-1}B \\ &= \begin{bmatrix} 0 & I \end{bmatrix} \begin{bmatrix} sI - A_{00} & -A_{01} \\ -K_1 & sI - K_2 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ I \end{bmatrix} \\ &= \left(sI - K_1(sI - A_{00})^{-1}A_{01} - K_2\right)^{-1}. \end{aligned} \tag{45}$$

Substituting $s = 0$ in (45), we have

$$R(0) = (K_1 A_{00}^{-1}A_{01} - K_2)^{-1} = \mathcal{Y}_2 > 0.$$

Hence, $R(s) + R(-s)^T$ must have full normal rank. Therefore, according to Lemma 2, the system (40) is SSNI.

Necessity: If A is Hurwitz, $R(s) + R(-s)^T$ has full normal rank and the system (40) is SSNI, then according to Lemma 2, there exists a matrix $Y = Y^T > 0$ such that $B = -AYC^T$ and $AY + YA^T < 0$.

Letting $X = Y^{-1}$, then $X = X^T > 0$. Also letting $Q = -(AY + YA^T)$, then we have $Q = Q^T > 0$ and $XA + A^TX = -XQX < 0$. Since $B = -AYC^T$, we have $CB + B^TC^T = -CAYC^T - CYA^TC^T = CQC^T$. Also, $XB - A^TC^T = -XAX^{-1}C^T - A^TC^T = -(XA + A^TX)X^{-1}C^T = XQXX^{-1}C^T = XQC^T$. Since $Q = Q^T > 0$, let $H := Q^{\frac{1}{2}}$. Hence, $H = H^T > 0$. We have

$$\begin{aligned} \begin{bmatrix} XA + A^TX & XB - A^TC^T \\ B^TX - CA & -(CB + B^TC^T) \end{bmatrix} &= -\begin{bmatrix} L^T \\ W^T \end{bmatrix} \begin{bmatrix} L & W \end{bmatrix} \\ &\leq 0 \end{aligned} \tag{46}$$

where $L = HX$ and $W = -HC^T$. Eq. (46) implies that for any $z \in \mathbb{R}^m$, $y \in \mathbb{R}^p$ and $v \in \mathbb{R}^p$, we have

$$\begin{bmatrix} z^T & y^T & v^T \end{bmatrix} \begin{bmatrix} XA + A^TX & XB - A^TC^T \\ B^TX - CA & -(CB + B^TC^T) \end{bmatrix} \begin{bmatrix} z \\ y \\ v \end{bmatrix}$$

$$= -\begin{bmatrix} z^T & y^T & v^T \end{bmatrix} \begin{bmatrix} L^T \\ W^T \end{bmatrix} \begin{bmatrix} L & W \end{bmatrix} \begin{bmatrix} z \\ y \\ v \end{bmatrix} \leq 0 \tag{47}$$

where equality holds if and only if $\begin{bmatrix} L & W \end{bmatrix} \begin{bmatrix} z \\ y \\ v \end{bmatrix} = 0$. That is $L\begin{bmatrix} z \\ y \end{bmatrix} + Wv = 0$, which is equivalent to $H\left(X\begin{bmatrix} z \\ y \end{bmatrix} - C^T v\right) = 0$. Because $H > 0$, this equation holds if and only if

$$X\begin{bmatrix} z \\ y \end{bmatrix} - C^T v = 0. \tag{48}$$

Let $X = \begin{bmatrix} X_{11} & X_{12} \\ X_{12}^T & X_{22} \end{bmatrix}$ and choose $y = 0$ and $v = -K_1 z$. With C given by (43), (48) becomes

$$\begin{bmatrix} X_{11} \\ X_{12}^T + K_1 \end{bmatrix} z = 0$$

which holds only if $X_{11}z = 0$. Since $X = X^T > 0$, $X_{11} = X_{11}^T > 0$. Hence, $X_{11}z = 0 \iff z = 0$. This implies that with the choice $y = 0$ and $v = -K_1 z$, strict inequality holds in (47) for all $z \neq 0$. Substituting (41)–(43) together with $y = 0$ and $v = -K_1 z$ into (47), we obtain

$$z^T(X_{11}A_{00} + A_{00}^T X_{11})z < 0$$

for all $z \neq 0$. This implies that $X_{11}A_{00} + A_{00}^T X_{11} < 0$. Therefore, A_{00} is Hurwitz. ■

We provide the following definition of the minimum phase property, which is needed in the presentation of the SSNI state feedback equivalence result.

Definition 12: (Minimum phase) [29], [44] A system (1) satisfying $rank(\mathcal{B}) = rank(\mathcal{C}) = p$ with relative degree vector $\{1, \ldots, 1\}$ is said to be minimum phase if its zero dynamics $\dot{z} = A_{00}z$ are asymptotically stable.

Theorem 3: Suppose the system (1) satisfying $rank(\mathcal{B}) = rank(\mathcal{C}) = p$ is minimal. Then, the following statements are equivalent.

1) The system has a relative degree vector $r = \{1, \ldots, 1\}$ and is minimum phase.
2) The system is state feedback equivalent to an SSNI system with realization (A, B, C), where A is Hurwitz, and the transfer function $R(s) := C(sI - A)^{-1}B$ is such that $R(s) + R(-s)^T$ has full normal rank.

Proof: The proof from Statements 1 to 2 follows directly from Lemmas 12 and 13. Note that the minimum phase condition is equivalent to the condition that A_{00} is Hurwitz in Lemma 13. Now, we prove that Statement 2 implies that the system has a relative degree vector $r = \{1, \ldots, 1\}$. SSNI systems form a subclass of all NI systems according to Definition 4. The analysis in the necessity proof of Theorem 1 also holds for SSNI systems except that strict inequalities hold for both (35) and (36), where this additional restriction comes from the strict inequality in Lemma 2. Strict inequality for (36) holds only if the zero block matrix has zero dimension, which is true only if $p_2 = 0$. This implies that Statement 2 is true only if the original system (1) with realization $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ can be output transformed by a nonsingular matrix $T_y \in \mathbb{R}^{p\times p}$ into a system with a relative degree vector $r = \{1, \ldots, 1\}$. According to Definition 5, i.e., the output transformed system satisfies $\det(\tilde{\mathcal{C}}\mathcal{B}) \neq 0$, where $\tilde{\mathcal{C}} = T_y\mathcal{C}$. Since $\det T_y \neq 0$, we have that $\det(\mathcal{C}\mathcal{B}) \neq 0$. This means that the original system (1) itself is already in a form with a relative degree vector $r = \{1, \ldots, 1\}$. Therefore, according to Lemma 12, (39) is the normal form of the system (1). The rest of the proof follows directly from Lemma 13. ■

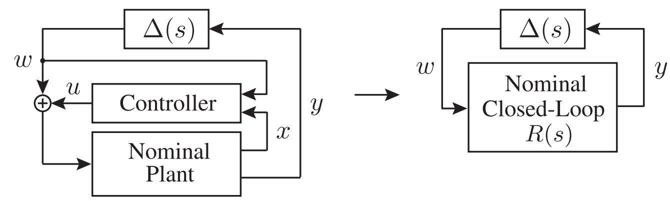

Fig. 1. Feedback control system. The plant uncertainty $\Delta(s)$ is SNI and satisfies $\lambda_{\max}(\Delta(0)) \leq \gamma$. Under some assumptions, we can find a controller such that the closed-loop transfer function $R(s)$ is NI with $R(\infty) = 0$ and $\lambda_{\max}(R(0)) < 1/\gamma$. Then the closed-loop system is aymptotically stable.

Remark 3: Note that SSNI systems belong to the class of OSNI systems and also the class of SNI systems (see [39]). Therefore, based on the control objective and conditions on the systems of interest, we can decide on whether to achieve the SSNI property or the OSNI property via the use of state feedback control.

V. Control of Systems With SNI Uncertainty

One useful application of state feedback equivalence to NI systems is to robustly stabilize systems for a class of uncertainties. More precisely, for a system having SNI uncertainty, we can render the nominal closed-loop system NI with the dc gain condition satisfied when full state measurement is available. A similar controller synthesis problem was investigated in [2], where the existence of such a state feedback controller depends on the solvability of a series of LMIs. However, in this article, the LMI conditions in [2] were replaced by some simpler conditions with respect to the relative degree vector and the weakly minimum phase property.

Consider the uncertain feedback control system in Fig. 1 and suppose that full state feedback is available. Then, Theorem 1 can be used in order to synthesize a state feedback controller such that the nominal closed-loop system is NI. Suppose the state-space model of the uncertain system in Fig. 1 is

$$\dot{x} = \mathcal{A}x + \mathcal{B}(u + w), \tag{49a}$$

$$y = \mathcal{C}x, \tag{49b}$$

$$w = \Delta(s)y \tag{49c}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^p$, and $y \in \mathbb{R}^p$ are the state, input, and output of the nominal plant. Here, (49c) models the uncertainty, and the uncertainty transfer function $\Delta(s)$ is assumed to be SNI satisfying $\lambda_{\max}(\Delta(0)) \leq \gamma$ for some constant $\gamma > 0$. Here, since $\Delta(s)$ is SNI, $\Delta(0)$ must be positive semidefinite.

The general idea used to stabilize the system (49) is to choose a control law u such that the system described by (49a) and (49b) is NI with input w and output y. Therefore, since $\Delta(s)$ is SNI, the system (49) forms a positive feedback interconnection of an NI system and an SNI system, whose equilibrium is asymptotically stable if the dc gain condition in Lemma 4 is satisfied.

Theorem 4: Consider the uncertain system (49). Suppose the realization $(\mathcal{A},\mathcal{B},\mathcal{C})$ is minimal with no zero at the origin. If there exists an output transformation $\tilde{y} = T_y y$, where $T_y \in \mathbb{R}^{p\times p}$ and $\det T_y \neq 0$, such that the realization $(\mathcal{A},\mathcal{B},T_y\mathcal{C})$ has relative degree less than or equal to two and is weakly minimum phase, then there exist $K_x \in \mathbb{R}^{p\times n}$ and $K_w \in \mathbb{R}^{p\times p}$ such that the control law

$$u = K_x x + K_w w$$

stabilizes the system (49).

Proof: According to Theorem 1 and its proof, the conditions here imply that the nominal plant in (49), described by

$$\dot{x} = \mathcal{A}x + \mathcal{B}u,$$
$$y = \mathcal{C}x$$

is state feedback equivalent to an NI system. Suppose the corresponding state feedback control law is

$$u = K_x x + K_v v. \tag{50}$$

Therefore, the nominal plant, described by (49a) and (49b), is NI with input w and output y under the control law

$$u = K_x x + K_w w \tag{51}$$

where

$$K_w = K_v - I. \tag{52}$$

Now, the system (49) is an interconnection of the nominal closed-loop NI system and the SNI uncertainty. To stabilize this interconnection, we investigate the dc gain conditions of Lemma 4. As is shown in the proof of Theorem 1, the output transformed system $(\mathcal{A},\mathcal{B},T_y\mathcal{C})$ is rendered NI with a transfer function $\widehat{R}(s)$ where $\widehat{R}(s) = C(sI-A)^{-1}B$ with A, B, and C given by (12), (13), and (14). We have that $\widehat{R}(\infty) = 0$. With the state feedback matrices given in the proof of Theorem 1, we also have that

$$\widehat{R}(0) = -CA^{-1}B = CA^{-1}AYC^T = CYC^T = \begin{bmatrix} \mathcal{Y}_2 & 0 \\ 0 & \mathcal{Y}_3 \end{bmatrix}$$

where we also use (30) and (32). Since the NI state feedback equivalence of the realization $(\mathcal{A},\mathcal{B},\mathcal{C})$ follows from the NI state feedback equivalence of the output transformed system $(\mathcal{A},\mathcal{B},T_y\mathcal{C})$, using Lemma 6, then the nominal closed-loop system can be rendered to be an NI system whose transfer function is $R(s) = T_y^{-1}\widehat{R}(s)T_y^{-T}$. Since $\mathcal{Y}_2$ and $\mathcal{Y}_3$ can be any positive definite matrices, we choose them to be such that

$$\lambda_{\max}\left(T_y^{-1}\begin{bmatrix} \mathcal{Y}_2 & 0 \\ 0 & \mathcal{Y}_3 \end{bmatrix}T_y^{-T}\right) < \frac{1}{\gamma}.$$

Therefore, $\lambda_{\max}(R(0)) < \frac{1}{\gamma}$. Hence, $\lambda_{\max}(R(0)\Delta(0)) < 1$. According to Lemma 4, it now follows that the system (49) is asymptotically stable. This completes the proof. ■

Remark 4: In the case that the uncertainty (49c) in the system (49) is NI, we can render the nominal closed-loop system (49a) and (49b) OSNI using the results of Theorem 2 in order to achieve asymptotic stability (see [8] and [26] for the corresponding OSNI stability results).

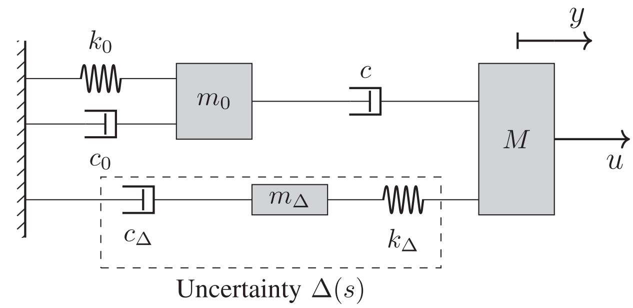

Fig. 2. Top view of a mass–spring-damper system containing uncertain parameters. The system consists of three masses, which move rectilinearly on a frictionless floor. The known parameters are $m_0 = M = 1\,kg$, $k_0 = 1\,N/m$ and $c_0 = c = 1Ns/m$ while the parameters $m\Delta$, k_Δ and c_Δ are unknown. A force input u is applied to the mass M and the displacement of M is measured as y.

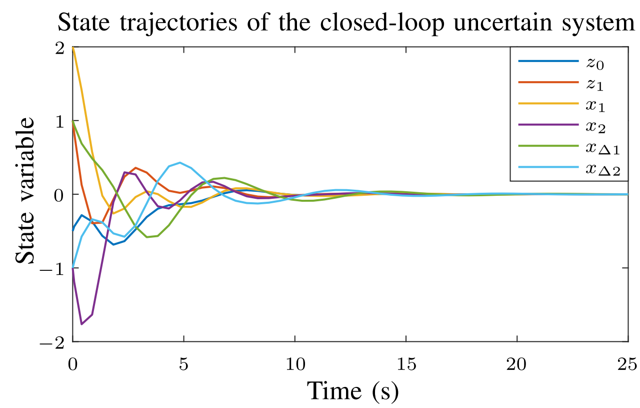

Fig. 3. State trajectory of the closed-loop system corresponding to the system shown in Fig. 2, which is also described by (57) and (53). Using the results in Theorems 1 and 4, we construct the state feedback control input (60). Under the input (60), all of the state variables converge to zero. This verifies that the system in Fig. 2 is asymptotically stabilized by the controller defined in (60).

VI. Illustrative Example

In this section, we illustrate our results provided in Section V using a real-world control example. As shown in Fig. 2, we consider three masses that slide on a frictionless floor. Mass m_0 is connected to a wall on its left by a spring k_0 and a damper c_0, and is connected to another mass M on its right by a damper c. The parameters m_0, k_0, c_0, M, and c are known; i.e., $m_0 = M = 1\,kg$, $k_0 = 1\,N/m$ and $c_0 = c = 1Ns/m$. We apply a force u to the mass M and measure its displacement y. A mass m_Δ is connected to M via a spring k_Δ and m_Δ is connected to the wall via a damper c_Δ. The parameters m_Δ, k_Δ, and c_Δ are unknown. This system can be regarded as arising in a vehicle suspension system (e.g., see [45]) or a vibration isolation system.

In Fig. 2, the part contained in the dashed rectangle can be regarded as the system uncertainty $\Delta(s)$. The uncertainty $\Delta(s)$ takes the displacement of M as its input u_Δ and produces a force output w, which is the force applied to M by the spring k_Δ.

Denoting the displacement of m_Δ by d, we obtain the equation of motion for the uncertainty $\Delta(s)$ using Newton's law as follows:

$$m_\Delta \ddot{d} = k_\Delta (y - d) - c_\Delta \dot{d}.$$

Letting $x_{\Delta 1} = d$ and $x_{\Delta 2} = \dot{x}_{\Delta 1}$, we obtain a state-space model for the uncertainty $\Delta(s)$ as

$$\dot{x}_{\Delta 1} = x_{\Delta 2}, \tag{53a}$$

$$\dot{x}_{\Delta 2} = -\frac{k_\Delta}{m_\Delta} x_{\Delta 1} - \frac{c_\Delta}{m_\Delta} x_{\Delta 2} + \frac{k_\Delta}{m_\Delta} u_\Delta, \tag{53b}$$

$$w = k_\Delta x_{\Delta 1} - k_\Delta u_\Delta. \tag{53c}$$

Therefore, the transfer function of the uncertainty $\Delta(s)$ given in (53) from input $u_\Delta = y$ to output w is

$$\Delta(s) = \frac{k_\Delta^2}{m_\Delta s^2 + c_\Delta s + k_\Delta} - k_\Delta. \tag{54}$$

Let z_0 denote the displacement of m_0. Using Newton's law, we obtain the following dynamic equations for the masses m_0 and M, respectively:

$$m_0 \ddot{z}_0 = c(\dot{y} - \dot{z}_0) - c_0 \dot{z}_0 - k_0 z_0; \tag{55}$$

$$M\ddot{y} = u + w + c(\dot{z}_0 - \dot{y}). \tag{56}$$

Using these two equations of motion, we can derive a state-space model for the system in Fig. 2 with input u and output y. Letting $z_1 = \dot{z}_0$, $x_1 = y$ and $x_2 = \dot{x}_1$, and substituting the values of m_0, M, k_0, c_0, c into (55) and (56), we have

$$\dot{z}_0 = z_1, \tag{57a}$$

$$\dot{z}_1 = -z_0 - 2z_1 + x_2, \tag{57b}$$

$$\dot{x}_1 = x_2, \tag{57c}$$

$$\dot{x}_2 = z_1 - x_2 + (u + w), \tag{57d}$$

$$y = x_1. \tag{57e}$$

This system is of the form (49a) and (49b). Hence, the system can be described by the left diagram in Fig. 1, where (57) is the nominal plant and (53) is the uncertainty $\Delta(s)$. We aim to apply the NI state feedback equivalence stabilization method in Theorem 4 to asymptotically stabilize this uncertain system. The system (57) is of relative degree two and is minimum phase since its internal dynamics described by (57a) and (57b) is asymptotically stable when $x_2 = 0$. We first make the system (57) NI by following the procedure given after Theorem 1. The system (57) is already in the form of (6). Indeed, (57a) and (57b) correspond to (6a). Eqs. (57c) and (57d) correspond to (6c) and (6d), respectively. Note that in this example, (6b) is irrelevant. For the system (57), the corresponding matrices in the normal form (6) are $A_{00} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$, $A_{02} = 0$, $A_{03} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $A_{30} = [0 \quad 1]$, $A_{32} = 0$, and $A_{33} = -1$. Since the matrix A_{00} is Hurwitz, then according to (18) and (20), we construct the matrix K_{20} using (21). Since A_{00} is Hurwitz, we can always find $\mathcal{Y}_1$ such that $A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T < 0$. For example, we choose $\mathcal{Y}_1 = \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix}$. Then, we have that $A_{00}\mathcal{Y}_1 + \mathcal{Y}_1 A_{00}^T = -\mathcal{Q} = \begin{bmatrix} -4 & 3 \\ 3 & -4 \end{bmatrix}$. Therefore, $\mathcal{Q} = \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix}$. According to (22), choose $\mathcal{H} = [0 \quad 0]$. Therefore, according to (21), we have

$$K_{20} = -A_{03}\mathcal{Y}_1^{-1} = \begin{bmatrix} -1 & -1.5 \end{bmatrix}.$$

The pair (A_{00}, K_{20}) is observable. Now, we construct the matrices K_{22} and K_{23} using (28) and (29), respectively. Since $A_{02} = 0$, we have $K_{22} = -\mathcal{Y}_3^{-1}$ and $K_{23} = -\frac{1}{2}I$, where $\mathcal{Y}_3 < 0$. Choosing $\mathcal{Y}_3 = 1$, then we have $K_{22} = -1$ and $K_{23} = -0.5$. According to (10) and (50), we construct the state feedback control input as

$$\begin{aligned} u &= v + (K_{20} - A_{30})z + (K_{22} - A_{32})x_1 + (K_{23} - A_{33})x_2 \\ &= v + \begin{bmatrix} -1 & -2.5 \end{bmatrix} z - x_1 - 0.5x_2 \end{aligned} \tag{58}$$

where $z = [z_0 \quad z_1]^T$. According to Theorem 1, the system (57) is made NI by the control input (58). Under the control input (58), the system (57) now becomes

$$\dot{z}_0 = z_1, \tag{59a}$$

$$\dot{z}_1 = -z_0 - 2z_1 + x_2, \tag{59b}$$

$$\dot{x}_1 = x_2, \tag{59c}$$

$$\dot{x}_2 = -z_0 - 1.5z_1 - x_1 - 0.5x_2 + v + w, \tag{59d}$$

$$y = x_1. \tag{59e}$$

Let $R(s)$ denote the transfer function of the system (59) from v to y. We have that

$$R(s) = \frac{(s+1)^2}{s^4 + 2.5\,s^3 + 4.5\,s^2 + 3.5s + 1}.$$

It can be verified that $R(s)$ satisfies Definition 1. We aim to let the system (57) have the form of (59) but with input w instead of v. Then, we implement a controller of the form (51) using (52) as described in the proof of Theorem 4. Based on (58), we obtain the corresponding control input

$$u = -z_0 - 2.5z_1 - x_1 - 0.5x_2 \tag{60}$$

since in this example $K_w = K_v - 1 = 0$. Now, the entire closed-loop system becomes the interconnection of $R(s)$ and $\Delta(s)$; i.e., $y = R(s)w$ and $w = \Delta(s)y$. We have that $R(\infty) = 0$, and $\Delta(0) = 0$ for any m_Δ, k_Δ, and c_Δ. Therefore, $\lambda(R(0)\Delta(0)) = 0 < 1$. The closed-loop interconnection of $R(s)$ and $\Delta(s)$ is asymptotically stable. Hence, the state feedback control input (60) asymptotically stabilizes the system shown in Fig. 2, which is described by (57) with $w = \Delta(s)y$, where $\Delta(s)$ is given in (54). This can also be verified via simulation. Since all of the parameters need to be known in the simulation, we let $m_\Delta = 0.5\,kg$, $k_\Delta = 0.5\,N/m$, and $c_\Delta = 0.1Ns/m$. Let the initial state variables of the system (57) and (53) be $[z_0(0) \quad z_1(0) \quad x_1(0) \quad x_2(0)]^T = [-0.5 \quad 1 \quad 2 \quad -1]^T$ and $[x_{\Delta 1}(0) \quad x_{\Delta 2}(0)]^T = [1 \quad -1]^T$, respectively. The state trajectories of the uncertain system shown in Fig. 2 are shown in Fig. 3. It can be observed that all of the state variables converge to zero.

VII. Conclusion

In this article, we have provided necessary and sufficient conditions under which a linear system can be rendered NI. As stated in Theorem 1, a minimal linear system (1) with no zeros at the origin is state feedback equivalent to an NI system if and only if it can be output transformed to a system, which has relative degree less than or equal to two and is weakly minimum phase. Similar OSNI and SSNI state feedback equivalence results are presented in Theorems 2 and 3. The NI state feedback equivalence results are then applied to robustly stabilize a system with SNI uncertainty. An example is also provided to illustrate the process of rendering a system NI in order to stabilize an uncertain system.

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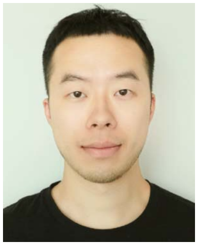

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